Convergence of Monte Carlo Optimistic Policy Iteration: Beyond Uniform State-Action Updates

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Abstract

The asymptotic behaviour of Monte Carlo optimistic policy iteration (MC-O-PI) is a long-standing open question. Classical convergence guarantees for general finite MDPs impose a restrictive sampling condition. In its canonical form, this condition requires that the episodes used for policy evaluation be initialised uniformly over the entire state-action space. This paper strictly relaxes that requirement. Specifically, we prove that initial-visit MC-O-PI converges to optimality even when updates are uniform only over the actions within each state. Different states may be sampled at unequal, adaptive frequencies, provided each accumulates infinite learning weight. With inertial policy ties and mild learning-rate regularity, this gives a realistic implementation when the state space is large or unknown but the action space in each state is manageable. The proof departs from the classical analysis of Tsitsiklis whose central commutativity argument no longer applies when states are updated at different frequencies. Instead, we first show that the mean-field dynamics of MC-O-PI generate monotonically improving policies when updates are uniform over the actions in each state, and then prove that noise cannot consistently prevent this improvement by extending the lock-in argument of the combined stability-ODE method. This approach suggests a new way to study optimistic policy-iteration algorithms in general.

Keywords: reinforcement learning, optimistic policy iteration, Monte Carlo control, convergence analysis, stability-ODE method

1 Introduction

Many reinforcement-learning algorithms follow the policy-iteration paradigm, which alternates between estimating the action values of a policy and improving the policy using those estimates. This framework is fundamental because it can learn an optimal policy for an unknown Markov decision process (MDP) using only sampled rewards and transitions. Classical policy iteration improves the policy once the action-value estimates have converged. In contrast, optimistic variants improve the policy after partial evaluation. This reduces computation as it requires fewer evaluation steps between each improvement step. However, it complicates the analysis because small perturbations to the value estimates can drastically affect the policy and alter the subsequent dynamics.

Convergence guarantees for optimistic policy-iteration algorithms remain limited, even in the tabular setting. Algorithms that use one-step bootstrapped estimates for evaluation, such

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as Q-learning, converge to optimality under standard asynchronous stochastic-approximation conditions. When using multi-step bootstrapped estimates, convergence to optimality requires additional structure, such as combining several multi-step approximations (Harutyunyan et al., 2016; Munos et al., 2016) or using a look-ahead mechanism (Winnicki and Srikant, 2023). Monte Carlo Optimistic Policy Iteration (MC-O-PI), which uses full-episode returns for evaluation, remains poorly understood. Despite its simplicity, the asymptotic behaviour of this foundational algorithm has been an open question for three decades (Sutton, 1999). Convergence to optimality is guaranteed when the MDP exhibits a feedforward structure (Wang et al., 2022; Lubars et al., 2021) or when the discount factor is smaller than 1/2 (Delattre and Fournier, 2025). The classical guarantees for general finite MDPs without a small-discount restriction require that all state-action pairs are updated at the same frequency (Tsitsiklis, 2002; Chen, 2018; Liu, 2021).

Updating all pairs as frequently requires that episodes are initialised uniformly over the entire state-action space. This condition is unrealistic in practice, especially when the state space is large. However, once the initial state of an episode is selected, uniformly sampling the initial action is usually straightforward. This paper proves that this weaker condition is sufficient. Specifically, initial-visit MC-O-PI converges almost surely to the optimal action values when updates are uniform over the actions within each state. Different states may thus be updated at arbitrary frequencies, provided each receives infinite accumulated learning. The precise sampling, policy-selection and learning-rate conditions are collected in Assumption 2. This result relaxes the state-sampling requirement of the previous convergence guarantees and applies in settings where uniform state sampling is not feasible but uniform action sampling is easy.

The proof of this new theorem, stated as Theorem 11 below, requires new tools. Previous work demonstrates convergence by studying the evolution of the value estimates. The arguments rely centrally on a commutativity property of the Bellman operator that fails when updates are not uniform over all state-action pairs. Instead, we track the evolution of the policy and find that the mean-field dynamics of MC-O-PI generate monotonically improving policies when updates are uniform over the actions within each state. We then show that stochastic perturbations cannot systematically defeat this improvement: at sufficiently late transitions, there is a fixed positive chance of improving the policy or of keeping the same action-value target forever. Boundedness allows this probability estimate to be made before exit from a fixed bounded set. Repeated opportunities then imply that only finitely many target changes occur, so the value estimates converge to the optimal action values.

This develops the combined stability–ODE method of Kushner and Yin (2003, Section 5.4) for entries into policy regions that may approach their boundaries. The proof explains how the learning-rate condition allows noise control even when these entry margins vanish.

This paper is organised as follows. Section 2 introduces finite Markov decision processes and formulates MC-O-PI as a stochastic difference inclusion. Section 3 proves the main convergence theorem, first for the mean-field dynamics and then for the stochastic algorithm. Section 4 discusses the practical implications of action-wise uniform updates and possible extensions to related optimistic methods. Unless stated otherwise, operations and relations

between functions are elementwise. For instance, given functions $f, g : \mathcal{X} \to \mathcal{Y}$ we write

$$(f+g)(x) = f(x) + g(x), \qquad \forall\, x \in \mathcal{X},$$
$$f \leq g \iff f(x) \leq g(x), \qquad \forall\, x \in \mathcal{X}.$$

Moreover, we compute norms as $\|f\| := \sup_{x \in \mathcal{X}} |f(x)|$ and adopt the convention $\inf \varnothing = \infty$.

2 Preliminaries

2.1 Finite Markov decision processes

A Markov decision process evolves over discrete steps. Throughout, the state set $\mathcal{S}$ and the action sets $\mathcal{A}(s)$ are finite and nonempty. We write $\mathcal{S} \times \mathcal{A}$ for the state-action pairs, with $a \in \mathcal{A}(s)$ understood. Terminal states have zero value and are omitted from $\mathcal{S}$; rewards have finite expected absolute value. At time t, the agent observes a state $s_t \in \mathcal{S}$ and selects an action $a_t \in \mathcal{A}(s_t)$ according to a policy $\pi(a_t|s_t)$. The environment then produces a reward r_t and transitions to a new state s_{t+1} according to its dynamics $p(s_{t+1}, r_t|s_t, a_t)$. If a policy π is deterministic in state s, then $\pi(s)$ directly refers to the action a where $\pi(a|s) = 1$. We denote the set of deterministic stationary policies by $\mathcal{P}$.

For any deterministic stationary policy π, the state-value function $v_\pi : \mathcal{S} \to \mathbb{R}$ measures the expected discounted return when following policy π starting in state s:

$$v_\pi(s) := \mathbb{E}_{p,\pi}\left[\sum_{t=0}^{\infty} \gamma^t\, r_t \;\middle|\; s_0 = s\right].$$

The action-value function $q_\pi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ measures the expected discounted return when taking action a in state s and then following policy π:

$$q_\pi(s,a) := \mathbb{E}_{p,\pi}\left[\sum_{t=0}^{\infty} \gamma^t\, r_t \;\middle|\; s_0 = s, a_0 = a\right].$$

A policy π is at least as good as another policy π' if $v_\pi \geq v_{\pi'}$, where the inequality holds at every state. This also implies $q_\pi \geq q_{\pi'}$ at every state-action pair. We denote this relation by $\pi \succeq \pi'$. If, additionally, there exists a state where $v_\pi(s) > v_{\pi'}(s)$, then π is better than π', denoted $\pi \succ \pi'$.

This paper considers discounted or episodic MDPs, meaning $\gamma \in [0, 1)$ or $\gamma = 1$ and episodes simulated with any deterministic stationary policy reach a terminal state almost surely. Such a policy is called *proper*. In a finite MDP, properness gives a geometric tail for the termination time and hence finite values (Bertsekas and Tsitsiklis, 1996, Section 2.2.1, Eqs. (2.7)–(2.9)). The Policy Improvement Theorem then allows us to compare policies using only the action-value function of one of them (Howard, 1960). We rely on a strict version of that theorem.

Lemma 1 *Given deterministic stationary policies π and π', suppose that*

$$q_\pi(s, \pi'(s)) \geq q_\pi(s, \pi(s)) \tag{1}$$

for every state $s \in \mathcal{S}$. Then $v_{\pi'} \geq v_\pi$. Moreover, if $q_{\pi'} \neq q_\pi$, then the improvement is strict, meaning there exists a state $s \in \mathcal{S}$ such that

$$v_{\pi'}(s) > v_\pi(s).$$

Proof. By the Policy Improvement Theorem, inequality (1) implies that $v_{\pi'} \geq v_\pi$. Suppose that $v_{\pi'} = v_\pi$. Then for every pair (s, a),

$$q_{\pi'}(s,a) = \mathbb{E}_p[r_0 + \gamma v_{\pi'}(s_1) \mid s_0 = s, a_0 = a] = \mathbb{E}_p[r_0 + \gamma v_\pi(s_1) \mid s_0 = s, a_0 = a] = q_\pi(s,a).$$

Thus, if $q_{\pi'} \neq q_\pi$, there exists at least one state where $v_{\pi'}(s) > v_\pi(s)$, and thus $\pi' \succ \pi$. ■

Classical results guarantee that, for discounted or episodic MDPs, there exists an optimal policy π_* that satisfies $q_{\pi_*}(s,a) = \max_\pi q_\pi(s,a)$ for all pairs (s,a). Moreover, π_* may be chosen to be deterministic and greedy with respect to q_{π_*}, meaning $\pi_*(s) \in \arg\max_{a \in \mathcal{A}(s)} q_{\pi_*}(s,a)$ for all s. The converse is also true: if a deterministic policy is greedy with respect to its own action values, then it must be optimal. This essential property allows us to verify whether a policy is optimal by inspecting its action values, rather than comparing it with all other policies.

2.2 Monte Carlo optimistic policy iteration

In many real-world applications, the transition dynamics and reward functions of the underlying MDP are unknown. Learning the optimal policy therefore requires interacting with the system to estimate expected returns from accumulated experience. One fundamental framework for this is *policy iteration*. This approach tracks an action-value estimate q and a policy π and refines them by alternately evaluating the current policy and improving it. This cycle is intended to converge to a policy that is greedy with respect to its own action values, and therefore optimal.

Monte Carlo optimistic policy iteration implements the policy-iteration framework by using *Monte Carlo policy evaluation*, which simulates complete episodes of the MDP to update q towards the action values of π, and *optimistic policy improvement*, which updates π to obtain higher returns as anticipated by q.

Specifically, consider the function space $\mathcal{Q} \coloneqq \{\mathcal{S} \times \mathcal{A} \to \mathbb{R}\}$. MC-O-PI starts from a finite deterministic estimate q_0 and generates a sequence $(q_k)_{k \geq 0}$ in $\mathcal{Q}$ as follows. At the beginning of iteration k, the algorithm selects a deterministic policy π_k that is greedy with respect to q_k:

$$\pi_k(s) \in \arg\max_{a \in \mathcal{A}(s)} q_k(s,a), \qquad \forall\, s \in \mathcal{S}. \tag{2}$$

The algorithm then samples an initial state-action pair (s_0, a_0) according to a sampling distribution σ_k, simulates an episode of the MDP following policy π_k, and observes the discounted return for each state-action pair in the episode. The observed return equals $q_{\pi_k}(s,a) + v_k(s,a)$, where $v_k(s,a)$ is a stochastic perturbation. Finally, the estimate q_k is updated at the chosen pairs appearing in the episode according to

$$q_{k+1}(s,a) = q_k(s,a) + \alpha_k(q_{\pi_k}(s,a) + v_k(s,a) - q_k(s,a)), \tag{3}$$

while all other estimates remain unchanged. Here $(\alpha_k)_{k\geq 0}$ is a scalar learning rate sequence that satisfies the Robbins-Monro conditions:

$$\sum_{k=0}^{\infty} \alpha_k = \infty, \qquad \sum_{k=0}^{\infty} \alpha_k^2 < \infty. \tag{4}$$

We use one global deterministic sequence of learning rates, with $0 < \alpha_k \leq 1$. The stochastic convergence proof also uses the following weaker form of eventual nonincrease: there are deterministic constants $c_\alpha \geq 1$ and $k_\alpha < \infty$ such that

$$\alpha_j \leq c_\alpha \alpha_k \qquad \text{for every } j \geq k \geq k_\alpha. \tag{5}$$

This permits occasional increases; for example, $\alpha_k = (2 + (-1)^k)/(k + 3)$ satisfies it with $c_\alpha = 3$.

Let $\mathcal{F}_k$ represent the available information after q_k is computed, but before selecting the policy, sampling distribution and update method for iteration k. Write $\mathcal{F}'_k$ for the information after these choices and before simulating the episode, so $\mathcal{F}_k \subseteq \mathcal{F}'_k \subseteq \mathcal{F}_{k+1}$. Conditional expectations given $\mathcal{F}'_k$ average over the remaining randomness in that episode. Any quantity already determined by this history can be taken outside the conditional expectation. We also use the tower property: averaging again over an earlier history gives the conditional expectation for that earlier history.

2.2.1 Greedy policy

Define the set-valued function $P : \mathcal{Q} \rightrightarrows \mathcal{P}$ that maps any action-value function q to the set of deterministic policies that are greedy with respect to q:

$$P(q) \coloneqq \big\{\pi \in \mathcal{P} : q(s, \pi(s)) = \max_{a \in \mathcal{A}(s)} q(s, a),\ \forall\, s \in \mathcal{S}\big\}. \tag{6}$$

At every step, the policy π_k thus satisfies

$$\pi_k \in P(q_k). \tag{7}$$

When several actions are greedy, we retain the previous action whenever it is still greedy:

$$\pi_{k+1}(s) = \pi_k(s) \quad \text{if } \pi_k(s) \in \arg\max_{a \in \mathcal{A}(s)} q_{k+1}(s, a), \qquad s \in \mathcal{S}. \tag{8}$$

The initial policy is any greedy policy, and when the previous action ceases to be greedy, any maximizing action may be selected. This inertia in the greedy policy selection means, in particular, that the policy does not change at a state that has not been updated.

2.2.2 Update function

Not all state-action pairs are necessarily updated after each episode. Given an episode $\big((s_t, a_t)\big)_{t\geq 0}$, there are three common update functions to decide which pairs to modify. The *initial-visit* method only updates the first state-action pair (s_0, a_0). The *first-visit* method updates each state-action pair (s_t, a_t) provided it has not appeared in the sequence

$\big((s_{t'}, a_{t'})\big)_{t'<t}$. The *every-visit* method updates all state-action pairs of an episode using every observed return from each pair.

Write f_k for the update method used at iteration k. For initial- or first-visit updates, let $u_k(s,a) = 1$ if the pair (s,a) is updated, and 0 otherwise. Set $v_k(s,a) = 0$ for pairs that are not updated. MC-O-PI then takes the form

$$q_{k+1}(s,a) = q_k(s,a) + \alpha_k u_k(s,a)\big(q_{\pi_k}(s,a) + v_k(s,a) - q_k(s,a)\big). \tag{9}$$

The indicator function u_k depends on the initial state-action pair of the episode, the policy used for simulating it and the chosen update method. Thus this representation separates the choice of updated pairs from the noise in their observed returns. The algebra below applies to either indicator-based update method; the convergence theorem in this paper uses initial visits.

For initial-visit updates, we assume that the sampled return is conditionally unbiased and has uniformly bounded conditional variance:

$$\mathbb{E}[v_k(s,a) \mid \mathcal{F}'_k, u_k(s,a) = 1] = 0, \qquad \mathbb{E}[v_k(s,a)^2 \mid \mathcal{F}'_k, u_k(s,a) = 1] \le c_v. \tag{10}$$

These conditions are imposed only when the pair has positive sampling probability. A fresh episode from the sampled pair, following π_k with the MDP transition and reward laws, suffices for conditional unbiasedness. If rewards are also bounded, the variance condition holds: discounting bounds the return directly, while the geometric termination tail of a proper finite-state policy gives a finite second moment. There are only finitely many deterministic policies and initial pairs, so the bound is uniform (Bertsekas and Tsitsiklis, 1996, Section 2.2.1).

2.2.3 Initial sampling distribution

To ensure that all state-action pairs are updated sufficiently often, we commonly impose *exploring starts* on the sampling distribution of the initial pair of each episode. This condition requires that every state-action pair has a non-zero probability of being selected:

$$\sigma_k(s,a) > 0 \qquad \text{for every } k \ge 0,\ s \in \mathcal{S},\ a \in \mathcal{A}(s).$$

However, we can effectively nullify this condition using quickly decaying sampling probabilities. For example, if $\sigma_k(s,a) \le 1/(k+2)^2$ for a particular pair, there is at least a 50% chance that no episode ever starts there. Under initial-visit updates, that pair would never be updated. A common sufficient condition is *strict exploring starts*: a fixed positive lower bound on every sampling probability. We instead require only infinite accumulated learning at each pair:

$$\sum_{k=0}^{\infty} \alpha_k \sigma_k(s,a) = \infty \quad \text{almost surely for every } s \in \mathcal{S},\ a \in \mathcal{A}(s). \tag{11}$$

This allows a pair to have zero sampling probability at some iterations, and it allows arbitrarily long waits between its updates. Infinite visitation alone would not suffice, since the learning rates at those visits could have a finite sum.

Our main result further requires that, conditional on the available history and selected initial state, the initial action is chosen uniformly from its available actions. We then write

$$\sigma_k(s,a) = \sigma_k(s) \geq 0, \qquad \sum_{s\in\mathcal{S}} |\mathcal{A}(s)|\sigma_k(s) = 1. \tag{12}$$

Here $\sigma_k(s)$ is the probability of each pair at state s, so the probability of choosing the state is $|\mathcal{A}(s)|\sigma_k(s)$. The state may be chosen adaptively using the available history, subject to (11).

2.2.4 Update distribution

The probability of updating each state-action pair after an episode depends on the distribution of the initial state-action pair, the policy used for simulating the episode and the update function. We gather these probabilities in the update distribution μ_k defined as

$$\mu_k := \mathbb{E}[u_k \mid \mathcal{F}'_k].$$

For initial-visit updates, $\mu_k = \sigma_k$. For first-visit updates, $\mu_k(s,a)$ is the probability that the pair appears anywhere in the episode; these probabilities need not sum to one.

This distribution controls the exploration-exploitation trade-off as it determines how frequently each action is updated. Exploration favours actions whose current estimated value is most uncertain, for instance by biasing μ_k towards actions that received few updates in the past. Exploitation, on the other hand, prioritises updating actions with the highest estimated value. It thus biases μ_k towards actions that are greedy with respect to q_k.

Using the update distribution, we can combine the stochastic terms u_k and v_k in equation (9) into a single term w_k defined as

$$w_k := u_k\, v_k + (u_k - \mu_k)(q_{\pi_k} - q_k). \tag{13}$$

This term incorporates the perturbations caused by the episode simulation and the asynchronous updates of state-action pairs. It simplifies equation (9) to

$$q_{k+1} = q_k + \alpha_k\big(\mu_k(q_{\pi_k} - q_k) + w_k\big). \tag{14}$$

2.2.5 Difference inclusion

After defining the sequences of learning rates $(\alpha_k)_{k\geq 0}$, start distributions $(\sigma_k)_{k\geq 0}$ and update functions $(f_k)_{k\geq 0}$, we can combine equations (7) and (14) into the difference inclusion

$$q_{k+1} \in \Big\{ q_k + \alpha_k\big(\mu_k(q_{\pi_k} - q_k) + w_k\big) : \\ \pi_k \in P(q_k), \mu_k = \mu(\sigma_k, \pi_k, f_k), w_k \sim w(q_k, \sigma_k, \pi_k, f_k) \Big\}. \tag{15}$$

Here $\mu(\sigma,\pi,f)$ and $w(q,\sigma,\pi,f)$ denote the conditional update probabilities and combined-noise law produced by the selected sampling distribution, policy and update method. Products of functions are evaluated separately at each state-action pair.

We gather the conditions for the main convergence result in one place.

Assumption 2 (Standing assumptions)

(1) ***Finite MDP and initialization.*** *The state and action sets are finite, rewards have finite expected absolute value, and the MDP is discounted or every deterministic stationary policy is proper. The initial estimate q_0 is finite and deterministic.*

(2) ***Learning rates.*** *The global deterministic rates satisfy $0 < \alpha_k \leq 1$, the Robbins–Monro conditions* (4)*, and the weakened monotonicity condition* (5)*.*

(3) ***Greedy policy selection.*** *The policies are greedy as in* (7)*, with inertia as in* (8)*.*

(4) ***Initial-visit updates.*** *Only the initial pair of each episode is updated. The sampled returns satisfy* (10) *for a deterministic constant $c_v < \infty$.*

(5) ***Sampling.*** *Conditional on the available history and initial state, the first action is uniform as in* (12)*, and every pair receives infinite accumulated learning as in* (11)*. The sampling distribution may depend on the available history and the selected policy.*

System (15) is a switched linear system whose active dynamics depend on which policy is greedy with respect to the current estimate q_k. Thus, as long as the greedy policy does not change, its action-value target remains the same, although the learning rates and update probabilities may vary. We can delimit the subsets of $\mathcal{Q}$ associated with each greedy policy using the set-valued function $Q : \mathcal{P} \rightrightarrows \mathcal{Q}$ defined as

$$Q(\pi) \coloneqq \big\{ q \in \mathcal{Q} : q(s, \pi(s)) = \max_{a \in \mathcal{A}(s)} q(s, a),\ \forall\, s \in \mathcal{S} \big\}. \tag{16}$$

This function is the inverse of P, defined in (6), as it maps any deterministic policy π to the set of action-value functions for which π is greedy. Since $Q(\pi)$ forms a convex subset of $\mathcal{Q}$, we refer to $Q(\pi)$ as the *region* of policy π in $\mathcal{Q}$.

2.2.6 Basic properties

The convergence proof needs two basic properties: bounded estimates, and convergence when the action-value target eventually stops changing. We first check that the combined perturbations have zero conditional mean and a controlled second moment. We then obtain boundedness by comparing the estimates with averages of returns having fixed lower and upper targets. The proofs are given in Appendices A.1–A.3.

Lemma 3 *Under Assumption 2, there exists a constant $c_w \in [0, \infty)$ such that for every time $k \geq 0$, every state $s \in \mathcal{S}$ and every action $a \in \mathcal{A}(s)$,*

$$\mathbb{E}\big[w_k(s, a) \mid \mathcal{F}'_k\big] = 0, \tag{17}$$

$$\mathbb{E}\big[w_k^2(s, a) \mid \mathcal{F}'_k\big] \leq c_w(1 + \|q_k\|^2). \tag{18}$$

Using Lemma 3 and the Robbins–Monro conditions, we can apply standard martingale and stochastic approximation arguments to establish fundamental properties of MC-O-PI.

Lemma 4 *Let $\mathcal{P}_\infty$ denote the set of policies that appear infinitely often in a sequence $(\pi_k)_{k\geq 0}$ generated by MC-O-PI. Under Assumption 2, the corresponding sequence $(q_k)_{k\geq 0}$ converges almost surely to the set*

$$\Big\{q \in \mathcal{Q} : \min_{\pi\in\mathcal{P}_\infty} q_\pi(s,a) \leq q(s,a) \leq \max_{\pi\in\mathcal{P}_\infty} q_\pi(s,a), \forall s \in \mathcal{S}, \forall a \in \mathcal{A}(s)\Big\}.$$

Convergence to this set means that the distance from q_k to it tends to zero. In particular,

$$\sup_{k\geq 0} \|q_k\| < \infty \qquad \text{almost surely.}$$

If the target action values in (15) were fixed, the same averaging argument would give convergence to that target. This conclusion follows directly from the return moments and accumulated learning; it does not require a separate bound on the estimates (Bertsekas and Tsitsiklis, 1996, Proposition 4.2). Thus, the asymptotic behaviour of MC-O-PI hinges on how its action-value targets evolve over time. To isolate these changes, we introduce *transition times*.

A random iteration is a stopping time when the history available at each iteration determines whether it has already occurred. For example, we can recognize a change of the current action-value target from the policies selected so far. All stopping times below use the histories $(\mathcal{F}'_k)$.

Definition 5 *Given a sequence of policies $(\pi_k)_{k\geq 0}$ and a deterministic starting iteration $r \geq 0$, define the transition times $(t_n)_{n\geq 0}$ as*

$$\begin{aligned} t_0 &:= r, \\ t_{n+1} &:= \inf\{k > t_n : q_{\pi_k} \neq q_{\pi_{t_n}}\}, \qquad t_n < \infty. \end{aligned}$$

We use $\inf \varnothing = \infty$ and set $t_{n+1} = \infty$ if $t_n = \infty$. For each finite t_n, the target stays constant at $q_{\pi_{t_n}}$ for all integer times $t_n \leq k < t_{n+1}$; if $t_{n+1} = \infty$, this means all $k \geq t_n$. Unless another starting iteration is specified, take $r = 0$.

Lemma 6 *Under Assumption 2, almost surely, if a finite transition time t_n satisfies $t_{n+1} = \infty$, then*

$$q_k \longrightarrow q_{\pi_{t_n}} = q_{\pi_*}.$$

Consequently, if $q_{\pi_{t_n}} \neq q_{\pi_*}$, then $t_{n+1} < \infty$ almost surely.

Proof. The fixed-target averaging argument in Appendix A.2 gives $q_k \to q_{\pi_{t_n}}$ on every path where the target becomes constant, outside one null event. Choose a policy that appears infinitely often after t_n. Its action-value function is $q_{\pi_{t_n}}$, and its greedy inequalities pass to the limit along that subsequence. It is therefore greedy with respect to its own action values and hence optimal. ■

Lemma 6 reduces the convergence problem to showing that only finitely many changes of the action-value target occur. The next section first establishes this for the mean-field dynamics, and then shows why stochastic perturbations cannot prevent it indefinitely.

3 Results

Experiments show that initial-visit MC-O-PI can converge to suboptimal solutions under arbitrary update distributions, even when the Robbins–Monro and strict exploring-starts conditions hold, meaning that every $\sigma_k(s,a)$ has a common positive lower bound. For instance, if at every step the actions associated with the current greedy policy are updated significantly less frequently than non-greedy actions, stable suboptimal equilibria can emerge even on the boundary of the optimal policy region. This section shows that uniform update distributions prevent such issues because the mean-field dynamics then generate monotonically improving policies. As a result, MC-O-PI converges to the optimal action values q_{π_*}.

This result extends the guarantees of Tsitsiklis (2002), Chen (2018) and Liu (2021) to a broader and more practical class of update distributions. Specifically, while they prove convergence to optimality when updates are *uniform over all state-action pairs*, we show that it remains true when updates are only *uniform over all actions in each state*.

Definition 7 (Uniform updates over all state-action pairs) *The update distribution μ is uniform over all state-action pairs if for all states $s, s' \in \mathcal{S}$ and actions $a \in \mathcal{A}(s)$ and $a' \in \mathcal{A}(s')$, we have*

$$\mu(s,a) = \mu(s',a').$$

Definition 8 (Uniform updates over actions in each state) *The update distribution μ is uniform over all actions in each state, if for every state $s \in \mathcal{S}$ and all actions $a, a' \in \mathcal{A}(s)$*

$$\mu(s,a) = \mu(s,a').$$

Definition 8 generalises Definition 7 because it allows updating different states at different frequencies. For instance, under initial-visit updates, the initial state of each episode may follow an adaptive distribution, provided every state accumulates infinite learning weight and the initial action in that state is sampled uniformly. This additional flexibility disrupts Tsitsiklis' proof as the essential commutation step fails (Tsitsiklis, 2002, p. 64).

This weaker condition is more practical for real-world applications, in particular when the state-space is large. For instance, chess has $O(10^{45})$ different board layouts but only $O(10^1)$ legal moves per layout. Uniformly sampling over the joint state-action space is thus not feasible, but sampling an action uniformly from a chosen state is straightforward. Henceforth, *uniform updates* refers to Definition 8 rather than Definition 7.

We first demonstrate that the mean-field dynamics under uniform updates generate monotonically improving policies. We then show that the stochastic perturbations in MC-O-PI cannot consistently prevent this improvement, and hence that MC-O-PI converges to the optimal action values. Since the non-contracting, discontinuous and non-convex MC-O-PI dynamics preclude using classical methods, we derive convergence from first principles.

3.1 Mean-field generates monotonically improving policies

Given that the stochastic perturbations in MC-O-PI are unbiased (Lemma 3), the mean-field update that arises when averaging out these perturbations is

$$q_{k+1} \in \left\{ q_k + \alpha_k \mu_k (q_{\pi_k} - q_k) : \pi_k \in P(q_k), \mu_k = \mu(\sigma_k, \pi_k, f_k) \right\} \tag{19}$$

with learning rates $(\alpha_k)_{k\geq 0}$, start distributions $(\sigma_k)_{k\geq 0}$ and update functions $(f_k)_{k\geq 0}$.

3.1.1 Policy improvement at every transition

Using Lemma 1, we show that policies generated by the mean-field dynamics in (19) improve monotonically because

$$q_{\pi_{t_n}}(s, \pi_{t_{n+1}}(s)) \geq q_{\pi_{t_n}}(s, \pi_{t_n}(s)) \tag{20}$$

for every state and at every transition.

Proposition 9 *Let $(\pi_k)_{k\geq 0}$ be a sequence of policies generated by the mean-field dynamics in* (19) *with uniform update distributions over all actions in each state (Definition 8). Let $(t_n)_{n\geq 0}$ be the associated transition times (Definition 5). Under Assumption 2, for every $n \in \mathbb{Z}_{\geq 0}$ with $t_{n+1} < \infty$, policy $\pi_{t_{n+1}}$ is better than π_{t_n}.*

Proof. Let $(q_k)_{k\geq 0}$ be a solution of (19) generated with update distributions $(\mu_k)_{k\geq 0}$, and let $(\pi_k)_{k\geq 0}$ be the resulting policy sequence with associated transition times $(t_n)_{n\geq 0}$.

Given policy π_{k+1} is greedy with respect to q_{k+1} for every time $k \geq 0$, we have $q_{k+1}(s, \pi_{k+1}(s)) \geq q_{k+1}(s, \pi_k(s))$ for every state $s \in \mathcal{S}$. Using (19), this becomes

$$\begin{aligned} q_k(s, \pi_{k+1}(s)) + \alpha_k \mu_k(s, \pi_{k+1}(s))\big(q_{\pi_k}(s, \pi_{k+1}(s)) - q_k(s, \pi_{k+1}(s))\big) \\ \geq q_k(s, \pi_k(s)) + \alpha_k \mu_k(s, \pi_k(s))\big(q_{\pi_k}(s, \pi_k(s)) - q_k(s, \pi_k(s))\big). \end{aligned}$$

Uniformity over the actions in each state gives $\mu_k(s, \pi_k(s)) = \mu_k(s, \pi_{k+1}(s)) =: \mu_k(s) \geq 0$. If $\mu_k(s) = 0$, no estimate changes at state s. The policy selection rule (8) therefore retains the previous action, so (21) holds with equality. If $\mu_k(s) > 0$, rearranging the previous inequality yields

$$q_{\pi_k}(s, \pi_{k+1}(s)) - q_{\pi_k}(s, \pi_k(s)) \geq \frac{1 - \alpha_k \mu_k(s)}{\alpha_k \mu_k(s)}\big(q_k(s, \pi_k(s)) - q_k(s, \pi_{k+1}(s))\big).$$

Since π_k is greedy with respect to q_k and, by assumption, $0 < \alpha_k \mu_k(s) \leq 1$, the right-hand side is nonnegative. As a result, we obtain

$$q_{\pi_k}(s, \pi_{k+1}(s)) \geq q_{\pi_k}(s, \pi_k(s)) \tag{21}$$

for every time $k \geq 0$ and every state $s \in \mathcal{S}$.

Applying the Policy Improvement Theorem to (21) gives $v_{\pi_{k+1}} \geq v_{\pi_k}$ at every step, and therefore $q_{\pi_{k+1}} \geq q_{\pi_k}$ as well. Since the target stays equal to $q_{\pi_{t_n}}$ for $t_n \leq k < t_{n+1}$, chaining (21) at each state also gives (20). By transitivity, $v_{\pi_{t_{n+1}}} \geq v_{\pi_{t_n}}$ at every finite transition. The two action-value functions differ by definition of a transition time. If their state-value functions were equal, the one-step expectation in the proof of Lemma 1 would make their action-value functions equal too. Thus the inequality is strict at some state, and $\pi_{t_{n+1}} \succ \pi_{t_n}$. ■

Proposition 9 indicates that MC-O-PI with uniform updates would visit improving policies if stochastic perturbations were averaged out, for instance by approximating q_{π_k} as the average return from many episodes.

3.1.2 Global convergence of mean-field solutions

The monotone policy improvement directly yields global convergence to q_{π_*}.

Proposition 10 *Solutions of the mean-field dynamics in* (19) *converge to the optimal action values q_{π_*} under Assumption 2, if the update distributions are uniform over all actions in each state (Definition 8).*

Technically, this result does not require all conditions in Assumption 2. It suffices that $0 < \alpha_k \leq 1$, the policy selection satisfies (8), and $\sum_k \alpha_k \mu_k(s) = \infty$ for every state s; square summability and (5) are not needed here. We discuss this relaxation in section 4.4.

3.2 Initial-visit MC-O-PI converges with uniform updates

This section proves that stochastic perturbations in initial-visit MC-O-PI cannot consistently offset the policy improvement of mean-field solutions. As a result, the algorithm converges to q_{π_*} when updates are uniform over the actions of each state.

Theorem 11 *Under Assumption 2, the action-value targets (q_{π_k}) change only finitely many times almost surely, and*

$$q_k \longrightarrow q_{\pi_*} \qquad \textit{almost surely.}$$

Every sufficiently late selected policy is optimal. If every state has a unique optimal action, the selected policy is eventually constant.

3.2.1 Proof strategy

The mean-field argument improves the policy at every transition. A sampled update can instead worsen it, so we seek a fixed positive probability of improvement at each sufficiently late transition. Since only finitely many deterministic policies exist, repeated opportunities for improvement will then imply convergence.

This follows the combined stability–ODE strategy of Kushner and Yin (2003, Section 5.4). The difficulty is that a trajectory can enter a policy region $Q(\pi)$ arbitrarily close to its boundary. Recurrence to that region alone does not give repeated visits to the fixed interior subset used in a standard lock-in estimate; see the setup in Borkar (2002, Section 3). We therefore first create a positive separation between the currently selected action and actions that would worsen the policy, and then control the probability that noise removes that separation. A sufficiently large fixed multiple of the current learning rate is enough: the separation may tend to zero at successive entries. Condition (5) makes the resulting probability bound uniform over sufficiently late entries. It is a sufficient condition for this proof, not a claimed necessity for convergence of the algorithm.

3.2.2 Actions that would worsen the policy

Consider a policy π selected at a time t. The pairs

$$\mathcal{B}_\pi = \big\{(s,a) \in \mathcal{S} \times \mathcal{A} : q_\pi(s,a) < q_\pi(s,\pi(s))\big\}$$

would worsen its selected action according to q_π. If none of these pairs becomes greedy, then at the next transition the new policy satisfies the Policy Improvement Theorem. To express this condition in terms of the estimates, we compare their estimated action values.

Definition 12 (Gap variables) *Fix a deterministic bound $K > \max_{\pi\in\mathcal{P}} \|q_\pi\|$ and a deterministic restart r. Let*

$$\zeta_r = \inf\{k \geq r : \|q_k\| > K\}$$

be the first time the estimate leaves this bound. For a stopping time $t \geq r$, on $\{t < \infty, t < \zeta_r\}$ set $\pi = \pi_t$ and let

$$\tau = \inf\{k > t : q_{\pi_k} \neq q_\pi\}$$

be the next change of the action-value target. For each pair $(s,a) \in \mathcal{B}_\pi$ and each $k \geq t$, define

$$d_k(s,a) = q_k(s,\pi(s)) - q_k(s,a), \tag{22}$$

$$d_\pi(s,a) = q_\pi(s,\pi(s)) - q_\pi(s,a). \tag{23}$$

The gap noise is $\xi_k(s,a) = w_k(s,\pi(s)) - w_k(s,a)$.

The target remains q_π before τ, even if the selected policy changes. Equal update probabilities within each state therefore give the gap recursion

$$d_{k+1}(s,a) = d_k(s,a) + \alpha_k\big(\sigma_k(s)(d_\pi(s,a) - d_k(s,a)) + \xi_k(s,a)\big). \tag{24}$$

Thus, on average, the estimate gap is pulled towards the positive target gap $d_\pi(s,a)$. The noise has the following bounds for $t \leq k < \tau \wedge \zeta_r$:

$$\mathbb{E}[\xi_k(s,a) \mid \mathcal{F}'_k] = 0, \qquad \mathbb{E}[\xi_k(s,a)^2 \mid \mathcal{F}'_k] \leq c_\xi \sigma_k(s), \qquad c_\xi = 2(c_v + 4K^2). \tag{25}$$

Indeed, only the initial pair is updated. The two possible updates forming this gap are disjoint, each has probability $\sigma_k(s)$, and before ζ_r the difference between an estimate and its target has magnitude at most $2K$. Their total second moment bounds the variance in (25). The stopping time ζ_r is used only to justify these deterministic bounds. Boundedness will remove it at the end of the proof; we never condition the noise on future containment.

We first determine how large a positive gap must be for stochastic updates to preserve it with high probability. We then construct that margin through favorable updates, a step called *seeding*, and combine the two estimates to obtain a positive probability of policy improvement.

Choose

$$\delta = \frac{1}{4} \min_{\substack{\pi\in\mathcal{P} \\ (s,a)\in\mathcal{B}_\pi}} d_\pi(s,a) > 0. \tag{26}$$

If the set in this minimum is empty, the Policy Improvement Theorem applies between every two policies in both directions. All policies then have the same optimal values, and Lemma 6 proves the theorem directly. We henceforth consider the nonempty case.

3.2.3 Preserving positive gaps

The following estimate bounds the probability that a positive gap closes before the next target change or exit. The drift and noise variance in (24)–(25) both carry the per-action update probability $\sigma_k(s)$ at state s. Measuring accumulated learning at that state will let us control this probability even when its updates are arbitrarily far apart. Write

$$\bar{\alpha}_k = \sup_{j\geq k} \alpha_j$$

for the largest remaining learning rate.

Lemma 13 (Gap preservation) *Consider a pair $(s,a) \in \mathcal{B}_\pi$ in Definition 12, and a stopping time $t' \geq t$ with $t' < \infty$, $t' \leq \tau \wedge \zeta_r$ and $d_{t'}(s,a) > 0$. Assume deterministic learning rates $0 < \alpha_k \leq 1$ with $\sum_k \alpha_k^2 < \infty$. Suppose that, under a given probability law, the gap recursion* (24) *and conditional moments* (25) *hold for $t' \leq k < \tau \wedge \zeta_r$. Under that law,*

$$\mathbb{P}\big(\exists \text{ finite } n \in [t', \tau \wedge \zeta_r] : d_n(s,a) \leq 0 \;\big|\; \mathcal{F}'_{t'}\big) \leq \frac{4c_\xi}{3\delta} \frac{\bar{\alpha}_{t'}}{d_{t'}(s,a)} + \frac{4c_\xi}{\delta^2} \sum_{k=t'}^{\infty} \alpha_k^2. \tag{27}$$

The event in (27) *includes the update ending at $\tau \wedge \zeta_r$.*

Equation (27) bounds the chance that one specified gap closes. If its starting value is bounded below by a fixed positive number, both terms become arbitrarily small as t' increases. If the starting gap also tends to zero, however, the first term depends on the ratio of the largest remaining learning rate to that gap: waiting alone need not make it small. A sufficiently large multiple of that learning rate controls this ratio, while a sufficiently late start makes the squared-learning-rate tail small. After proving the estimate, we construct such a margin through favorable updates.

Proof. Fix (s,a) and write $d_k = d_k(s,a)$ within this proof. Stop the noise sum at the first time the gap d_n reaches zero or below, or at the endpoint $\tau \wedge \zeta_r$, whichever comes first:

$$\eta = (\tau \wedge \zeta_r) \wedge \inf\{n \geq t' : d_n \leq 0\},$$
$$M_n = \sum_{k=t'}^{n-1} \mathbf{1}_{\{k<\eta\}} \alpha_k \xi_k(s,a).$$

Define the accumulated learning at state s by

$$A_n = \sum_{k=t'}^{n-1} \alpha_k \sigma_k(s). \tag{28}$$

We distinguish whether the gap d_k reaches or exceeds δ before it first becomes nonpositive.

The gap d_n reaches zero or below before reaching δ.

We consider the event that $d_n \leq 0$ at a finite time $n \in [t', \tau \wedge \zeta_r]$, with $0 < d_k < \delta$ at every preceding time $t' \leq k < n$. Thus reaching zero includes equality, and reaching the upper level δ means reaching or exceeding it. We first determine how negative the accumulated noise M_n must be to bring the gap d_n to zero or below, then bound the probability of that noise fluctuation.

For $k \geq t'$, recall that the stopped noise sum M_n retains increment k exactly when $k < \eta$, where η is the first time the gap becomes nonpositive or the endpoint $\tau \wedge \zeta_r$ is reached. The indicator $\mathbf{1}_{\{k<\eta\}}$ and the learning rate α_k are known from $\mathcal{F}'_k$. Condition (25) applies when $k < \eta$; when $k \geq \eta$, the stopped increment is zero. Substituting the noise increment $\alpha_k \xi_k(s,a)$ and taking these known factors outside the conditional expectation, the zero-mean condition (25) gives

$$\begin{aligned} \mathbb{E}[M_{k+1} - M_k \mid \mathcal{F}'_k] &= \mathbb{E}[\mathbf{1}_{\{k<\eta\}} \alpha_k \xi_k(s,a) \mid \mathcal{F}'_k] \\ &= \mathbf{1}_{\{k<\eta\}} \alpha_k \mathbb{E}[\xi_k(s,a) \mid \mathcal{F}'_k] \\ &= 0. \end{aligned} \tag{29}$$

Because the conditional mean in (29) is zero, the conditional variance is the conditional second moment. Squaring an indicator leaves it unchanged. We again take the known factors outside the conditional expectation and apply the variance bound (25). For the last step, use $\alpha_k \le \bar{\alpha}_{t'}$ and the accumulated-learning identity $A_{k+1} - A_k = \alpha_k \sigma_k(s)$ from (28). For $k \ge t'$, this gives

$$\begin{aligned}
\mathbb{E}[(M_{k+1} - M_k)^2 \mid \mathcal{F}'_k] &= \mathbb{E}[\mathbf{1}_{\{k<\eta\}} \alpha_k^2 \xi_k(s,a)^2 \mid \mathcal{F}'_k] \\
&= \mathbf{1}_{\{k<\eta\}} \alpha_k^2 \mathbb{E}[\xi_k(s,a)^2 \mid \mathcal{F}'_k] \\
&\le \mathbf{1}_{\{k<\eta\}} c_\xi \alpha_k^2 \sigma_k(s) \\
&\le c_\xi \alpha_k^2 \sigma_k(s) \\
&\le c_\xi \bar{\alpha}_{t'} (A_{k+1} - A_k).
\end{aligned} \tag{30}$$

Suppose now that the gap first becomes nonpositive at n, without previously reaching δ. For $t' \le k < n$, the gap satisfies $0 < d_k < \delta$, whereas (26) gives the target gap $d_\pi(s,a) \ge 4\delta$. The mean-field drift term $\alpha_k \sigma_k(s)(d_\pi(s,a) - d_k)$ in (24) is therefore at least $3\delta \alpha_k \sigma_k(s)$. Summing (24), with every noise increment up to this first nonpositive gap retained in M_n, gives

$$0 \ge d_n \ge d_{t'} + 3\delta A_n + M_n. \tag{31}$$

Thus the accumulated noise M_n must offset both the starting gap $d_{t'}$ and the accumulated mean-field increase $3\delta A_n$.

Write b_n for their sum. To control the probability that the noise M_n falls to $-b_n$ or below, rescale each noise increment by the corresponding threshold:

$$\begin{aligned}
b_n &= d_{t'} + 3\delta A_n, \\
Z_n &= \sum_{k=t'}^{n-1} \frac{M_{k+1} - M_k}{b_{k+1}}.
\end{aligned} \tag{32}$$

We will relate a downward crossing of $-b_n$ by M_n to a fixed level for the rescaled noise $|Z_n|$.

We consider $n > t'$, as is necessarily the case when the positive starting gap $d_{t'}$ first becomes nonpositive. Both noise sums start at zero: $M_{t'} = Z_{t'} = 0$. By (32), $M_{k+1} - M_k = b_{k+1}(Z_{k+1} - Z_k)$. When we expand and sum these differences, each Z_k with $t' < k < n$ appears twice: once multiplied by b_k, and once subtracted with multiplier b_{k+1}. The term involving $Z_{t'}$ vanishes, leaving the endpoint term $b_n Z_n$. This is summation by parts, and it gives

$$\begin{aligned}
M_n &= \sum_{k=t'}^{n-1} b_{k+1}(Z_{k+1} - Z_k) \\
&= \sum_{k=t'+1}^{n} b_k Z_k - \sum_{k=t'}^{n-1} b_{k+1} Z_k \\
&= b_n Z_n - \sum_{k=t'+1}^{n-1} (b_{k+1} - b_k) Z_k.
\end{aligned} \tag{33}$$

The thresholds b_k are positive and nondecreasing because the accumulated learning A_k is nondecreasing. Hence the weights $b_{k+1} - b_k$ are nonnegative and sum to $b_n - b_{t'+1} \le b_n$.

Applying the triangle inequality to (33), then bounding each $|Z_k|$ by its largest value up to n, yields

$$\begin{aligned} |M_n| &\le b_n|Z_n| + \sum_{k=t'+1}^{n-1} (b_{k+1}-b_k)|Z_k| \\ &\le \left[b_n + \sum_{k=t'+1}^{n-1} (b_{k+1}-b_k) \right] \max_{t'\le m\le n} |Z_m| \\ &\le 2b_n \max_{t'\le m\le n} |Z_m|. \end{aligned} \tag{34}$$

If the gap d_n reaches zero or below before reaching δ, (31) requires $M_n \le -b_n$. Equation (34) then forces $|Z_m| \ge 1/2$ at some finite time $m \in [t', n]$. Consequently,

$$\begin{aligned} &\mathbb{P}\left(\begin{array}{l} \exists \text{ finite } n \in [t', \tau\wedge\zeta_r] : d_n \le 0, \\ 0 < d_k < \delta \text{ for every } t' \le k < n \end{array} \;\middle|\; \mathcal{F}'_{t'} \right) \\ &\qquad \le \mathbb{P}\big(\exists \text{ finite } m \ge t' : |Z_m| \ge \tfrac{1}{2} \,\big|\, \mathcal{F}'_{t'}\big)\,. \end{aligned} \tag{35}$$

To bound the probability on the right of (35), we use Doob's martingale maximal inequality. It bounds the probability that a martingale reaches a given magnitude by its expected squared terminal value divided by the square of that magnitude. We justify its application here before estimating the resulting second moments.

Doob's inequality requires square integrability. The denominator $b_{k+1} = b_k + 3\delta\alpha_k\sigma_k(s)$ is known from $\mathcal{F}'_k$, so dividing the increment in (29) by b_{k+1} preserves its conditional zero mean whenever the rescaled increment is integrable. However, the positive starting gap $d_{t'}$ may be arbitrarily small, so integrability needs checking.

For this check only, fix $\varepsilon > 0$ and multiply the process Z_n by $\mathbf{1}_{\{d_{t'}\ge\varepsilon\}}$. This leaves the noise sum Z_n unchanged on histories with $d_{t'} \ge \varepsilon$, and sets it to zero on the remaining histories. The indicator is known at the restart. Since $b_{k+1} \ge d_{t'}$, equation (30) gives

$$\begin{aligned} &\mathbb{E}\Big[\big(\mathbf{1}_{\{d_{t'}\ge\varepsilon\}}(Z_{k+1}-Z_k)\big)^2 \,\Big|\, \mathcal{F}'_k \Big] \\ &\qquad \le \frac{1}{\varepsilon^2}\mathbb{E}[(M_{k+1}-M_k)^2 \mid \mathcal{F}'_k] \\ &\qquad \le \frac{c_\xi \alpha_k^2}{\varepsilon^2}. \end{aligned} \tag{36}$$

The modified increments have conditional mean zero, and (36), together with $\sum_k \alpha_k^2 < \infty$, makes the modified noise sum square-integrable.

First suppose the restart t' is a fixed finite time, and take a finite horizon $n \ge t'$. Apply Doob's inequality at magnitude $1/2$ to this square-integrable noise sum. The conditional version follows by applying the ordinary inequality after multiplying the noise sum by the indicator of any event known at the restart. Also, distinct zero-mean increments have zero expected product: when the later increment is drawn, the earlier one is already known and the later one has conditional mean zero. Thus the expected squared terminal value equals the expected sum of squared increments. On histories with $d_{t'} \ge \varepsilon$, the modified noise sum

equals Z_n, and we obtain

$$\begin{aligned}\mathbb{P}\Big(\max_{t'\le m\le n}|Z_m|\ge\tfrac{1}{2}\,\Big|\,\mathcal{F}'_{t'}\Big)&\\ \le\frac{\mathbb{E}[Z_n^2\mid\mathcal{F}'_{t'}]}{(1/2)^2}&\\ =4\,\mathbb{E}\left[\sum_{k=t'}^{n-1}(Z_{k+1}-Z_k)^2\,\middle|\,\mathcal{F}'_{t'}\right].&\end{aligned}\tag{37}$$

If the restart t' is random, apply this argument separately on $\{t'=j\}$. This event is known at time j, so the same indicator argument applies there; combining the countably many finite values of j gives (37) at the random restart.

We can now remove the temporary cutoff. For each $\varepsilon=1,1/2,1/3,\ldots$, (37) holds on $\{d_{t'}\ge\varepsilon\}$. These events cover every positive starting gap $d_{t'}$, so (37) holds for every such gap, outside one null event. The finite second-moment bound (36) need not be uniform as ε decreases: the inequality (37) that it justifies contains no ε. No subsequent calculation is restricted to a fixed positive lower bound on $d_{t'}$.

Finally increase the finite horizon n. The events on the left of (37) increase to the event that $|Z_m|$ reaches $1/2$ at some finite time, while the nonnegative sums on the right increase to the full sum. Monotone convergence therefore gives

$$\begin{aligned}\mathbb{P}\big(\exists\text{ finite }m\ge t':|Z_m|\ge\tfrac{1}{2}\,\big|\,\mathcal{F}'_{t'}\big)&\\ \le 4\,\mathbb{E}\left[\sum_{k\ge t'}(Z_{k+1}-Z_k)^2\,\middle|\,\mathcal{F}'_{t'}\right].&\end{aligned}\tag{38}$$

Together, (35) and (38) explain why bounding the squared increments of the rescaled noise Z_n bounds the probability that the gap d_n reaches zero or below before reaching δ.

The denominator b_{k+1} is known from $\mathcal{F}'_k$, so it can be taken outside the conditional expectation. Substituting (30) into the definition (32) yields

$$\begin{aligned}\mathbb{E}[(Z_{k+1}-Z_k)^2\mid\mathcal{F}'_k]&=\frac{\mathbb{E}[(M_{k+1}-M_k)^2\mid\mathcal{F}'_k]}{b_{k+1}^2}\\ &\le\frac{c_\xi\bar\alpha_{t'}(A_{k+1}-A_k)}{(d_{t'}+3\delta A_{k+1})^2}.\end{aligned}\tag{39}$$

The tower property lets us average these conditional second-moment bounds at the starting history $\mathcal{F}'_{t'}$ and sum them in (38).

To estimate the resulting sum, the function $x\mapsto(d_{t'}+3\delta x)^{-2}$ is decreasing. A rectangle with width $A_{k+1}-A_k$ and height at the right endpoint A_{k+1} lies below its graph. Therefore

$$\frac{A_{k+1}-A_k}{(d_{t'}+3\delta A_{k+1})^2}\le\int_{A_k}^{A_{k+1}}\frac{\mathrm{d}x}{(d_{t'}+3\delta x)^2}.\tag{40}$$

The intervals $[A_k,A_{k+1}]$ are consecutive and start at $A_{t'}=0$. Extending them to $[0,\infty)$ gives an upper bound even if the accumulated learning A_n has a finite limit. The integral depends only on the starting gap $d_{t'}$, so its conditional expectation given $\mathcal{F}'_{t'}$ is unchanged.

Combining (35), (38), (39) and (40), the conditional probability that the gap d_n reaches zero or below before reaching δ, by time $\tau \wedge \zeta_r$, is at most

$$\begin{aligned} & 4\,\mathbb{E}\left[\sum_{k\geq t'} \frac{c_\xi \bar{\alpha}_{t'}(A_{k+1}-A_k)}{(d_{t'}+3\delta A_{k+1})^2}\,\middle|\,\mathcal{F}'_{t'}\right] \\ & \quad \leq 4c_\xi \bar{\alpha}_{t'} \int_0^\infty \frac{\mathrm{d}x}{(d_{t'}+3\delta x)^2} \\ & \quad = \frac{4c_\xi \bar{\alpha}_{t'}}{3\delta d_{t'}}. \end{aligned} \tag{41}$$

This bound applies even when the positive starting gaps $d_{t'}$ tend to zero at successive entries. Its size depends on the ratio $\bar{\alpha}_{t'}/d_{t'}$; seeding will ensure that the starting gaps are sufficiently large relative to the remaining learning rates. It remains to bound the probability that the gap d_n reaches zero or below after first reaching δ.

The gap d_n reaches zero or below after reaching δ. Suppose that the first nonpositive gap occurs at a finite time $n \leq \tau \wedge \zeta_r$, and that $d_k \geq \delta$ at some preceding time $t' \leq k < n$. Let ℓ be the last such time. We first show that falling from $d_\ell \geq \delta$ to $d_n \leq 0$ requires a noise fluctuation of at least δ.

In the update at ℓ, the noise-free part is a convex combination of the gap $d_\ell \geq \delta$ and the target gap $d_\pi(s,a) \geq 4\delta$, since $0 \leq \alpha_\ell \sigma_\ell(s) \leq 1$. It is therefore at least δ. Equation (24) gives

$$\begin{aligned} d_{\ell+1} &= (1-\alpha_\ell\sigma_\ell(s))d_\ell + \alpha_\ell\sigma_\ell(s)d_\pi(s,a) + (M_{\ell+1}-M_\ell) \\ &\geq \delta + M_{\ell+1} - M_\ell. \end{aligned} \tag{42}$$

At every subsequent time $\ell < k < n$, the gap satisfies $0 < d_k < \delta$ by the choices of n and ℓ. The mean-field drift in (24) is then nonnegative. Summing the remaining updates and using (42), we obtain

$$0 \geq d_n \geq \delta + M_n - M_\ell. \tag{43}$$

Thus $M_n - M_\ell \leq -\delta$. By the triangle inequality,

$$\delta \leq |M_n - M_\ell| \leq |M_n| + |M_\ell| \leq 2 \max_{t' \leq m \leq n} |M_m|. \tag{44}$$

Consequently, the stopped noise sum M_m must reach magnitude $\delta/2$ at some finite time. The last-visit index ℓ is used only in these pathwise inequalities; conditioning remains at the stopping time t'.

By (30) and $\sigma_k(s) \leq 1$, the conditional second moments of the increments of M_m are at most $c_\xi \alpha_k^2$. Their sum is finite, so M_m is square-integrable without a cutoff on the starting gap $d_{t'}$. Applying Doob's inequality at magnitude $\delta/2$, with the conditional and infinite-horizon steps justified above, bounds the probability of this second case by

$$\begin{aligned} & \mathbb{P}\big(\exists \text{ finite } m \geq t' : |M_m| \geq \tfrac{\delta}{2} \,\big|\, \mathcal{F}'_{t'}\big) \\ & \quad \leq \frac{4}{\delta^2}\,\mathbb{E}\left[\sum_{k\geq t'} (M_{k+1}-M_k)^2 \,\middle|\, \mathcal{F}'_{t'}\right] \\ & \quad \leq \frac{4c_\xi}{\delta^2} \sum_{k\geq t'} \alpha_k^2. \end{aligned} \tag{45}$$

Both cases retain the noise increment ending at $\tau \wedge \zeta_r$. Adding the bounds (41) and (45) proves (27). ■

3.2.4 Seed: creating positive gaps

The preservation bound (27) now gives us a concrete objective: lower-bound its denominator $d_{t'}(s,a)$ relative to the largest remaining learning rate. At a newly selected policy, some estimated gaps may be zero. We create the required margin through a finite number of favorable updates. States may wait arbitrarily long between updates, so each state completes this construction separately; its gaps will then be protected while other states are still waiting.

Suppose the reference action $\pi(s)$ is still selected and $d_k(s,a) < \delta$. Since

$$\big(q_\pi(s,\pi(s)) - q_k(s,\pi(s))\big) + \big(q_k(s,a) - q_\pi(s,a)\big) = d_\pi(s,a) - d_k(s,a) > 3\delta,$$

at least one of these two estimation errors exceeds $3\delta/2$. We request a favorable update as follows:

1. If $q_\pi(s,\pi(s)) - q_k(s,\pi(s)) > 3\delta/2$, update $(s,\pi(s))$ and require $v_k(s,\pi(s)) \geq -\delta/2$.
2. Otherwise update (s,a) and require $v_k(s,a) \leq \delta/2$; in this case $q_k(s,a) - q_\pi(s,a) > 3\delta/2$.

Either update increases the intended gap by more than $\delta\alpha_k$, leaves every other gap against the reference action unchanged or larger, and keeps the reference action greedy. The selection rule (8) therefore retains it.

The conditional return moments ensure that each one-sided noise requirement has probability at least $\delta^2/(4c_v + \delta^2)$, by the one-sided Chebyshev inequality (see Appendix A.5 for the conditional calculation). Conditional on selecting the state, each action is sampled uniformly. Thus every favorable update has conditional probability at least

$$p_0 = \frac{\delta^2}{\max_{s\in\mathcal{S}} |\mathcal{A}(s)|\,(4c_v + \delta^2)} > 0. \tag{46}$$

No lower bound on the probability of selecting the state is needed.

Definition 14 (Seed construction) *For the gaps and stopping times in Definition 12, fix a positive integer N. To give every gap an initial margin, order the actions in $\mathcal{B}_\pi$ at each state. Skip a pair if its gap has already reached δ; otherwise request up to N favorable updates for it, stopping earlier if it reaches δ. Whenever an unfinished state is sampled, request the next favorable update on its list. Completed states receive ordinary updates. Stop the construction at the first failed request or at $\tau \wedge \zeta_r$. Write T_s for the iteration when state s completes its list, with $T_s = t$ if it is complete initially and $T_s = \infty$ if the construction stops first. Skips are processed at t and after every successful update, before the next sample, so T_s is a stopping time.*

Lemma 15 (Seed) *Under Assumption 2, consider the gaps and stopping times of Definition 12, with $t \geq k_\alpha$, and let N be a positive integer. The seed construction of Definition 14*

requests at most $|\mathcal{S}\times\mathcal{A}|N$ favorable updates. Along successful requests, each unfinished state retains its reference action. If $T_s<\infty$, then

$$d_{T_s}(s,a)\geq \min\{\delta, N\delta\bar{\alpha}_{T_s}/c_\alpha\} \qquad \text{for every } a \text{ with } (s,a)\in\mathcal{B}_\pi. \tag{47}$$

This includes a state completed at the endpoint $\tau\wedge\zeta_r$. Each request has conditional probability at least p_0 given the current history and the selection of its unfinished state.

Proof. A favorable update only raises the reference estimate or lowers a competing estimate. It keeps the reference action greedy, so (8) retains that action. At a state not updated, the policy is unchanged. Induction therefore retains the reference action at every unfinished state, including during waits and after its last favorable update.

Every processed gap either reaches δ or receives N increases of more than $\delta\alpha_k$. Later favorable updates at the same state cannot lower it. For each seeding step $k<T_s$, condition (5) gives $\alpha_j\leq c_\alpha\alpha_k$ for every $j\geq T_s$, hence $\bar{\alpha}_{T_s}\leq c_\alpha\alpha_k$. Summing these N increases proves (47). At most $|\mathcal{S}\times\mathcal{A}|$ pairs are processed, and the probability of each request is bounded below by (46). ■

3.2.5 A positive probability of policy improvement

The seed lemma supplies the margin required by the gap-preservation bound. We combine them by choosing the number of favorable updates first, and then a sufficiently late start. This gives a fixed positive probability that the next target change improves the policy, unless the estimates first leave the bound or the target stays constant forever.

Lemma 16 *Under Assumption 2, fix $K>\max_{\pi\in\mathcal{P}}\|q_\pi\|$. There are deterministic constants $k_0<\infty$ and $p>0$ such that, for every deterministic restart r and stopping time $t\geq\max\{r,k_0\}$, the times in Definition 12 satisfy*

$$\mathbb{P}\big(\tau<\zeta_r,\ \pi_\tau\not\succ\pi_t \;\big|\; \mathcal{F}'_t\big)\leq 1-p \tag{48}$$

on $\{t<\infty, t<\zeta_r\}$. Thus the stopped interval either reaches the bound, continues forever with the same target, or ends with a better policy, with conditional probability at least p.

Proof. Choose a positive integer N such that

$$\frac{4|\mathcal{S}\times\mathcal{A}|c_\xi c_\alpha}{3\delta^2 N}\leq\frac{1}{4}.$$

Then choose a deterministic $k_0\geq k_\alpha$ so that

$$N\bar{\alpha}_{k_0}\leq c_\alpha, \qquad \frac{4|\mathcal{S}\times\mathcal{A}|c_\xi}{\delta^2}\sum_{k=k_0}^{\infty}\alpha_k^2\leq\frac{1}{4}. \tag{49}$$

These choices are possible because $\bar{\alpha}_k\to 0$ and the squared learning rates have a finite sum.

The guarantee that all seed requests succeed is only $p_0^{|\mathcal{S}\times\mathcal{A}|N}$, whereas the gap-loss bound decreases as $1/N$. Subtracting the latter bound from the former does not give a positive success probability. We need to control gap loss on runs where seeding succeeds.

Simply conditioning on successful seeding can bias the later updates at completed states, because those updates can affect when the construction stops. We therefore assign different probabilities to the same possible runs, only for this proof. Given the current history, keep the state-selection probabilities unchanged. If the selected state is unfinished, condition its requested action and return on being favorable; at completed states and after the construction stops, keep the original conditional action and return law.

Write $\widehat{\mathbb{P}}$ for these auxiliary probabilities. Under this law every requested update succeeds. The construction in Lemma 17 in Appendix A.6 justifies these conditional laws and shows that, for every future event E,

$$\mathbb{P}(E \mid \mathcal{F}'_t) \geq p_0^{|\mathcal{S}\times\mathcal{A}|N}\widehat{\mathbb{P}}(E \mid \mathcal{F}'_t) \quad \text{on } \{t < \infty, t < \zeta_r\}. \tag{50}$$

Thus a lower bound under the auxiliary probabilities gives a lower bound for the actual algorithm. The factor accounts for at most $|\mathcal{S}\times\mathcal{A}|N$ favorable requests, each with conditional probability at least p_0.

For a state completing at $T_s < \tau \wedge \zeta_r$, its subsequent update law is unchanged. The gap recursion and conditional moments therefore hold under this law, and Lemma 13 applies starting at T_s. Equations (47) and (49) give

$$\frac{\bar{\alpha}_{T_s}}{d_{T_s}(s,a)} \leq \max\left\{\frac{\bar{\alpha}_{T_s}}{\delta}, \frac{c_\alpha}{N\delta}\right\} \leq \frac{c_\alpha}{N\delta},$$

since $T_s \geq t \geq k_0$. Thus the same bound covers gaps that reached δ and those built by N favorable updates. The conditional probability of losing this gap through the endpoint is consequently at most

$$\frac{4c_\xi c_\alpha}{3\delta^2 N} + \frac{4c_\xi}{\delta^2}\sum_{k=t}^{\infty}\alpha_k^2. \tag{51}$$

Multiply the conditional estimate by the event $\{T_s < \tau \wedge \zeta_r\}$, which is known at T_s, and take conditional expectations given $\mathcal{F}'_t$. Summing over at most $|\mathcal{S}\times\mathcal{A}|$ pairs bounds the chance of any completed gap closing by $1/2$. No independence or simultaneous completion is required.

On the complementary event, a finite target transition before exit cannot select a pair in $\mathcal{B}_\pi$. States that completed earlier have positive gaps; states still unfinished retain the reference action, as do states completed exactly at the endpoint. Hence

$$q_\pi(s, \pi_\tau(s)) \geq q_\pi(s, \pi(s)) \qquad \text{for every state } s \in \mathcal{S}.$$

The two targets differ at a transition, so Lemma 1 gives $\pi_\tau \succ \pi$. We do not need every state to complete: at a finite endpoint unfinished states are safe, while an infinite endpoint without exit already gives a constant target. Transferring back with (50) proves (48) with

$$p = \frac{1}{2}p_0^{|\mathcal{S}\times\mathcal{A}|N} > 0.$$

If $\mathcal{B}_\pi$ is empty for the reference policy, the policy-improvement inequality holds for every next selected policy, so the same bound applies without a seed construction. An exit after an unsuccessful finite transition is not counted as success for that earlier interval. ■

3.2.6 Global convergence of MC-O-PI

The preceding result gives a uniform chance of progress on each continuing interval. We now repeat these opportunities. There can be at most $|\mathcal{P}| - 1$ consecutive strict policy improvements, so a sufficiently long run of successful intervals must end either with exit from the fixed bound or with a target that stays constant. Boundedness then removes the exit alternative.

Proof of Theorem 11. Fix $K > \max_{\pi \in \mathcal{P}} \|q_\pi\|$, and take k_0, p from Lemma 16. For a deterministic restart $r \geq k_0$, use the transition times of Definition 5 with starting iteration r. Stop counting immediately if $\|q_r\| > K$, and subsequently if exit occurs before or at the next target change, or if the target stays constant forever without exit. Every finite continuing transition time is a stopping time.

At each continuing interval, stopping or strict policy improvement has conditional probability at least p. By induction and the tower property, stopping within the next m intervals or obtaining m consecutive improvements has conditional probability at least p^m. On a branch already stopped, the continuation probability is one; on a finite successful-transition branch, condition at its stopping time and apply the bound again. No independence is used, and we never continue by conditioning at an infinite time.

There cannot be $|\mathcal{P}|$ consecutive strict improvements. Thus, from every continuing interval, the conditional probability of stopping within the next $|\mathcal{P}|$ intervals is at least $p^{|\mathcal{P}|}$. Applying this bound in successive blocks gives probability at most

$$\left(1 - p^{|\mathcal{P}|}\right)^j$$

of continuing through j blocks. This tends to zero, so infinitely many target changes before ζ_r have probability zero. Unsuccessful transitions may worsen the policy; the bound still holds because it concerns the chance of a new run of consecutive successful intervals from each surviving block.

Intersect these probability-one conclusions over the deterministic integers $r \geq k_0$. By Lemma 4, almost every path remains inside the bound K after some such integer. On that path, choose r with $\zeta_r = \infty$. It follows that only finitely many target changes occur. The restart is chosen pathwise after the countable intersection, without treating eventual containment as a stopping time.

By Lemma 6, $q_k \to q_{\pi_*}$ almost surely. At each state, every nonoptimal action has a positive gap below the optimal action value. Finiteness and convergence exclude those actions from all sufficiently late greedy selections. Every sufficiently late policy is therefore optimal. If each state has a unique optimal action, the policy is eventually constant. ■

The proof thus follows a policy-level argument: monotone mean-field improvement, a positive chance that stochastic updates preserve it, and finitely many possible successive improvements. The stochastic estimates handle small gaps at policy transitions and arbitrarily long waits between state updates.

4 Discussion

Theorem 11 showed that initial-visit MC-O-PI converges almost surely to q_{π_*} when updates are uniform over the actions in each state. This section explores the implications of this

result. We first explain that the uniformity condition effectively requires using initial-visit updates, which does not necessarily slow down convergence compared to first-visit updates. We then discuss possible extensions to temporal-difference updates and to estimates stored approximately at every step. Finally, we discuss the roles of cumulative learning, step-size regularity and policy ties.

4.1 Uniform updates effectively force initial-visits

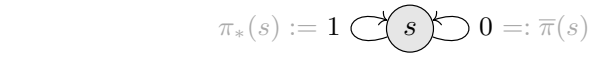
For first-visit updates, $\mu_k(s,a)$ is the probability that pair (s,a) appears in the episode and is updated. These probabilities need not sum to one. For every-visit updates, the corresponding expected update counts can also exceed one. We call these quantities *update rates*. When the underlying MDP is unknown, both depend on its transition dynamics under π_k. Therefore, requiring uniform update rates effectively forces the use of initial-visit updates, because by updating only the initial state-action pair of each episode, we ensure μ_k equals σ_k and thus retain control.

This control comes at the cost of data efficiency: the initial-visit method discards within-episode transitions that the first-visit method would reuse. However, more updates per episode do not automatically translate into faster policy convergence. The extra updates induce unequal action update rates that can generate sliding modes in the action-value estimates $(q_k)_{k\geq 0}$ and thereby slow down the convergence of the policy sequence $(\pi_k)_{k\geq 0}$. Ultimately, it is the convergence speed of this policy sequence that truly matters for identifying the optimal behaviour of the MDP.

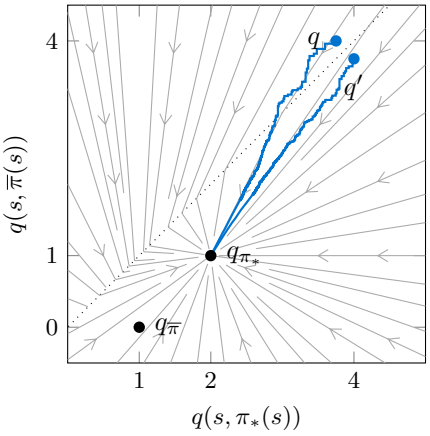
To illustrate this, consider the MDP in Figure 1a with a discount factor $\gamma = 0.5$. Action *left* yields a reward of one and *right* yields a reward of zero. Under a uniform sampling distribution σ_k, the intial-visit method updates exactly one state-action pair per episode, whereas the first-visit method updates on average 1.5 pairs. With a learning rate $\alpha_k = 1/(k+100)$, the first-visit method also benefits from larger effective steps because it updates state-action pairs more frequently.

Despite this apparent advantage, the initial-visit method is more stable than first-visit when MC-O-PI is initialised above q_{π_*}. Namely, initial-visit solutions follow straight lines that always point towards better policies. In contrast, first-visit solutions exhibit a sliding mode that causes high-frequency oscillations in the policy sequence and traps q_k near the boundary for longer. As a result, first-visit can take longer to settle on the optimal policy. For instance, when averaging over 1000 solutions starting in q', first-visit ones require on average 43 updates to settle on π_*, whereas initial-visit ones always remain in $Q(\pi_*)$. When starting in q, the solutions take 68 and 70 updates to settle on π_*.

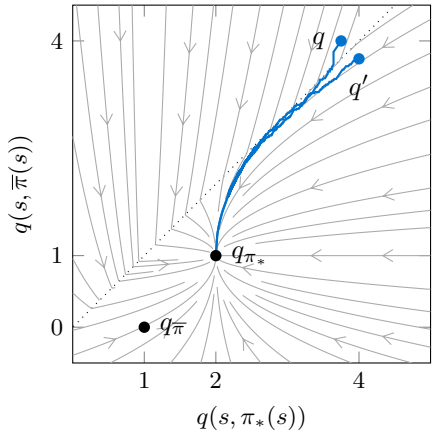
Overall, this experiment illustrates that the initial-visit method can be practical in real-world applications. By avoiding sliding modes, it sometimes converges to the optimal policy π_* as fast as, or faster than, the first-visit method despite fewer updates and smaller effective learning rates. Besides, the initial-visit method is more reliable for early stopping. Namely, if training stops early, the last greedy policy of an initial-visit solution is typically the best policy encountered so far, whereas the first-visit solution may be oscillating across a decision boundary, making it unclear which policy is best.



(a) MDP



(b) Initial-visit solutions



(c) First-visit solutions

Figure 1: By avoiding sliding modes, initial-visit solutions sometimes converge to the optimal policy π_* faster than first-visit solutions despite fewer updates and smaller effective learning rates. The blue lines are MC-O-PI solutions starting in q or q', whereas the grey lines indicate the mean-field flow.

4.2 Mean-field analysis with temporal difference updates

The gap estimate uses conditional drift and variance bounds, while the favorable-update construction uses a bounded number of requests regardless of the waiting time. An analogous approach to temporal-difference updates would additionally need a policy-improvement mechanism and a favorable-update construction compatible with targets that change with the current estimate. The present theorem does not establish those properties. The additional difficulty would be to understand how the estimate q and the simulation depth n affect the target $T_\pi^n q$, where $\pi \in P(q)$. Here T_π is the action-value Bellman operator,

$$(T_\pi q)(s,a) = \mathbb{E}_p[r_0 + \gamma q(s_1, \pi(s_1)) \mid s_0 = s, a_0 = a]\,,$$

and the superscript n denotes repeated composition.

4.3 MC-O-PI with function approximation

Theorem 11 applies specifically to *tabular* MC-O-PI, where the estimate q_k can be stored exactly. In practical applications, however, the estimate is approximated, for instance using a neural network. In this setting, the error between the observed return and the current estimate is then used to update the parameters of the approximation.

Extending the theorem would require quantitative control of accumulated approximation errors and a replacement for the favorable-update argument for individual state-action pairs. Representational richness alone does not provide either property. Perturbation analyses such

as Bertsekas and Tsitsiklis (1996, Section 4.3.3) and Kushner and Yin (2003, Section 5.2) suggest possible tools, but their applicability to this extension remains to be established.

Finally, recent breakthroughs in natural language processing and robotics (Schulman et al., 2017) rely on policy gradient methods. Wagner (2013) shows that some of these methods can be interpreted as optimistic policy iteration algorithms with function approximation. Extending our results to this setting might thus improve the efficiency and performance of these essential methods.

4.4 Sampling, learning rates and policy ties

The theorem requires infinite accumulated learning at every state, $\sum_k \alpha_k \sigma_k(s) = \infty$, but no positive lower sampling bound and no bound on the waiting time between updates. This distinction matters because the algorithm uses one global deterministic learning-rate sequence: infinitely many visits can carry only finite total weight. The deterministic argument needs only the divergent effective weights and coefficients in $[0, 1]$; square summability is used to control stochastic errors.

Inertia in greedy policy selection, as in (8), is convenient for the proof. Fixed action priorities give the same convergence conclusion, provided the priorities are used consistently from initialization. The only change is the local retention argument: raising the reference estimate, lowering a competitor, or leaving a state's table unchanged cannot introduce a higher-priority maximizer. Thus unfinished states retain their reference actions during the favorable-update construction, and the rest of the proof is unchanged. Fresh random tie-breaking would require a different retention argument and is not included in the theorem as stated.

The learning-rate condition bounds future increases by a fixed factor; it allows infinitely many increases and imposes no comparability lower bound on future steps. In the seed bound, it compares each favorable update directly with the largest learning rate after completion. The gap-preservation estimate then controls future noise relative to the resulting gap without using the condition again. Whether this regularity can be removed from the stochastic theorem remains open here. Its role in the proof does not establish its necessity for convergence.

5 Conclusion

We established that the mean-field dynamics of Monte Carlo optimistic policy iteration drive trajectories toward monotonically improving policies when updates are uniform over all actions in each state. We then proved that the stochastic perturbations cannot systematically counteract this improvement, and consequently that MC-O-PI converges to the optimal action values when updates are uniform.

This result extends existing convergence guarantees to more practical settings. Unlike previous proofs that required uniform updates across all state-action pairs, our analysis allows different states to be updated at different frequencies. This relaxation makes the algorithm applicable to large-scale problems where uniform sampling over the joint state-action space is not feasible. The guarantee allows such state-dependent sampling under the stated cumulative-learning, return-moment, tie-breaking and learning-rate conditions.

We extended the lock-in strategy of the combined stability-ODE method to accommodate the discontinuous dynamics of MC-O-PI. This mean-field approach provided a more intuitive explanation for the convergence mechanism than previous proofs that directly tackled the stochastic process. More importantly, it opens up new possibilities, beyond contraction-based analysis, to study the convergence of optimistic policy iteration algorithms.

We see three principal directions for future work: (1) determine whether the remaining step-size regularity can be weakened further, (2) apply this new method to study the convergence of optimistic policy iteration algorithms that rely on temporal difference evaluation, and (3) generalise our result when value estimates are stored approximately.

Appendix A. Proofs of auxiliary results

A.1 Proof of Lemma 3

Fix a time $k \geq 0$, a state $s \in \mathcal{S}$ and an action $a \in \mathcal{A}(s)$.

By definition,

$$w_k(s,a) = u_k(s,a)v_k(s,a) + \big(u_k(s,a) - \mu_k(s,a)\big)\big(q_{\pi_k}(s,a) - q_k(s,a)\big). \tag{52}$$

By the return-moment assumption (10), and $u_k(s,a)v_k(s,a) = 0$ on $\{u_k(s,a) = 0\}$, it follows that

$$\mathbb{E}[u_k(s,a)v_k(s,a) \mid \mathcal{F}'_k] = \mathbb{E}\big[\mathbf{1}_{\{u_k(s,a)=1\}}\mathbb{E}[v_k(s,a) \mid \mathcal{F}'_k, u_k(s,a) = 1] \mid \mathcal{F}'_k\big] = 0. \tag{53}$$

Also, $\mathbb{E}[u_k(s,a) \mid \mathcal{F}'_k] = \mu_k(s,a)$, and μ_k and $q_{\pi_k} - q_k$ are $\mathcal{F}'_k$-measurable. Hence,

$$\begin{aligned}\mathbb{E}\big[\big(u_k(s,a) - \mu_k(s,a)\big)\big(q_{\pi_k}(s,a) - q_k(s,a)\big) \mid \mathcal{F}'_k\big] \\ = \big(q_{\pi_k}(s,a) - q_k(s,a)\big)\big(\mathbb{E}\big[u_k(s,a) \mid \mathcal{F}'_k\big] - \mu_k(s,a)\big) = 0.\end{aligned} \tag{54}$$

Summing (53) and (54) yields $\mathbb{E}[w_k(s,a) \mid \mathcal{F}'_k] = 0$. This proves (17).

Finally, since $u_k^2(s,a) \leq 1$ and $(u_k(s,a) - \mu_k(s,a))^2 \leq 1$, using $(x+y)^2 \leq 2x^2 + 2y^2$ on (52) yields

$$\begin{aligned}w_k^2(s,a) &\leq 2v_k^2(s,a) + 2(q_{\pi_k}(s,a) - q_k(s,a))^2 \\ &\leq 2v_k^2(s,a) + 2(2q_{\pi_k}^2(s,a) + 2q_k^2(s,a)).\end{aligned}$$

Since all action-values are bounded, there exists a constant $c_{\mathcal{P}} < \infty$ such that $\|q_\pi\|^2 \leq c_{\mathcal{P}}$ for every policy $\pi \in \mathcal{P}$. As a result,

$$\begin{aligned}\mathbb{E}\big[w_k^2(s,a) \mid \mathcal{F}'_k\big] &\leq 2c_v + 2\,\mathbb{E}\big[(q_{\pi_k}(s,a) - q_k(s,a))^2 \mid \mathcal{F}'_k\big] \\ &\leq 2c_v + 4c_{\mathcal{P}} + 4\|q_k\|^2\end{aligned}$$

So (18) holds for example with $c_w := \max\{2c_v + 4c_{\mathcal{P}}, 4\}$. ∎

A.2 Fixed-target averages

The comparison argument for boundedness and the convergence of a constant-target block both use the following consequence of the supermartingale convergence theorem (Bertsekas and Tsitsiklis, 1996, Proposition 4.2): if a nonnegative process has conditional expected increments bounded above by a negative term plus a summable error, then it converges and the negative terms have finite sum.

Fix a deterministic restart r, a policy $\pi \in \mathcal{P}$ and a state-action pair (s,a). Starting from $x_r = q_r(s,a)$, consider the average with fixed target $q_\pi(s,a)$,

$$x_{k+1} = x_k + \alpha_k u_k(s,a)\big(q_\pi(s,a) + v_k(s,a) - x_k\big).$$

The initial squared error is integrable: finite-time estimates have finite second moments by induction in (9) and (10). Since $u_k(s,a)$ is an indicator with conditional mean $\sigma_k(s,a)$, the return moments and $0 < \alpha_k \leq 1$ give

$$\mathbb{E}\big[(x_{k+1} - q_\pi(s,a))^2 \mid \mathcal{F}'_k\big] \leq (1 - \alpha_k\sigma_k(s,a))(x_k - q_\pi(s,a))^2 + c_v\alpha_k^2.$$

The cited theorem shows that $(x_k - q_\pi(s,a))^2$ converges and that

$$\sum_{k=r}^{\infty} \alpha_k \sigma_k(s,a)(x_k - q_\pi(s,a))^2 < \infty \quad \text{almost surely.}$$

Its limit must be zero, since (11) makes the sum of the weights infinite. These assertions hold simultaneously for every deterministic integer r, every policy and every state-action pair: take the intersection of their countably many probability-one events. We may therefore choose a restart and a target separately on each resulting path, without conditioning on their future-dependent values.

A.3 Proof of Lemma 4

Let (q_k) and (π_k) be an MC-O-PI solution satisfying Assumption 2, and work on the probability-one event just obtained. Since the policy set is finite, there is a finite index r after which only policies in $\mathcal{P}_\infty$ appear. Thus for all $k \geq r$ and every pair (s,a),

$$\min_{\pi \in \mathcal{P}_\infty} q_\pi(s,a) \leq q_{\pi_k}(s,a) \leq \max_{\pi \in \mathcal{P}_\infty} q_\pi(s,a).$$

Start $x_r = y_r = q_r(s,a)$ and use the same sampled updates and return noise, with the two fixed targets:

$$x_{k+1} = x_k + \alpha_k u_k(s,a) \left(\min_{\pi \in \mathcal{P}_\infty} q_\pi(s,a) + v_k(s,a) - x_k \right),$$
$$y_{k+1} = y_k + \alpha_k u_k(s,a) \left(\max_{\pi \in \mathcal{P}_\infty} q_\pi(s,a) + v_k(s,a) - y_k \right).$$

Because $0 \leq \alpha_k u_k(s,a) \leq 1$, induction gives $x_k \leq q_k(s,a) \leq y_k$ for every $k \geq r$. Each comparison target is one of the finitely many values covered by the preceding fixed-target calculation. Consequently,

$$\liminf_{k \to \infty} q_k(s,a) \geq \min_{\pi \in \mathcal{P}_\infty} q_\pi(s,a),$$
$$\limsup_{k \to \infty} q_k(s,a) \leq \max_{\pi \in \mathcal{P}_\infty} q_\pi(s,a).$$

This proves convergence to the stated bounded set. In particular, the estimates are bounded along almost every path. If the actual target is eventually constant, the estimate itself is one of the fixed-target averages above, which also justifies its use in Lemma 6. ■

A.4 Proof of Proposition 10

Let (q_k) be a solution of the mean-field dynamics (19) under the stated assumptions, with policies (π_k) and transition times (t_n). Proposition 9 shows that their action-value functions are nondecreasing and change only by improvements. Since there are finitely many deterministic policies, the target is constant from some finite transition time t_n onwards. For every pair (s,a) and every $k \geq t_n$, unrolling the deterministic recursion gives

$$q_k(s,a) - q_{\pi_{t_n}}(s,a) = \big(q_{t_n}(s,a) - q_{\pi_{t_n}}(s,a)\big) \prod_{j=t_n}^{k-1} (1 - \alpha_j \mu_j(s,a)).$$

The product tends to zero because $\sum_j \alpha_j \mu_j(s,a) = \infty$. A policy appearing infinitely often in the remaining sequence has action values $q_{\pi_{t_n}}$; passing its greedy inequalities to the limit shows that it is greedy for its own action values, hence optimal. Therefore $q_{\pi_{t_n}} = q_{\pi_*}$ and $q_k \to q_{\pi_*}$. ■

A.5 The one-sided return bound used in Lemma 15

For a zero-mean random variable X with $\mathbb{E}[X^2] \le c_v$, the nonnegative quantity $(X + 2c_v/\delta)^2$ exceeds $(\delta/2 + 2c_v/\delta)^2$ on $\{X > \delta/2\}$. Taking expectations gives

$$\mathbb{P}(X > \delta/2) \le \frac{c_v + (2c_v/\delta)^2}{(\delta/2 + 2c_v/\delta)^2} = \frac{4c_v}{4c_v + \delta^2}.$$

Taking complements yields $\mathbb{P}(X \le \delta/2) \ge \delta^2/(4c_v + \delta^2)$; applying the same argument to $-X$ gives $\mathbb{P}(X \ge -\delta/2) \ge \delta^2/(4c_v + \delta^2)$. The calculation also holds conditionally, so (10) supplies both return bounds used in the seed stage.

A.6 Probabilities during the seed construction

Lemma 17 *Under the conditions of Lemma 15, there is an auxiliary probability law $\widehat{\mathbb{P}}$ under which every requested seed update succeeds. State-selection probabilities are unchanged, and at completed states the action and return law given the current history and selected state is unchanged. The same holds after the construction stops. Every event of original probability zero still has auxiliary probability zero, and, for every future event E,*

$$\mathbb{P}(E \mid \mathcal{F}'_t) \ge p_0^{|\mathcal{S}\times\mathcal{A}|N} \widehat{\mathbb{P}}(E \mid \mathcal{F}'_t) \quad \text{on } \{t < \infty, t < \zeta_r\}.$$

Proof. We construct the auxiliary law by assigning new probabilities to the same algorithm runs. At each requested update, discard the failed outcomes and divide the probabilities of successful outcomes by their total conditional probability. This is ordinary conditioning at that update; the weights below keep track of these changes over the whole run. At a requested update of state s, let $p_k(s)$ be the conditional probability of the favorable action and return event, given $\mathcal{F}'_k$ and that state s is selected. By (46), $p_k(s) \ge p_0$ on positive-probability selections. Values at zero-probability selections may be assigned arbitrarily.

Starting with $L_t = 1$, multiply L_k by the indicator that the requested update succeeds, divided by $p_k(s)$. At iterations without a request, use factor one. Each factor has conditional mean one given the current history and selected state. Thus

$$\mathbb{E}[L_{k+1} \mid \mathcal{F}'_k] = L_k.$$

This is the martingale property: knowing the present history does not change the expected future weight. At most $|\mathcal{S} \times \mathcal{A}|N$ factors differ from one, so

$$0 \le L_k \le p_0^{-|\mathcal{S}\times\mathcal{A}|N}.$$

The weight is eventually constant on each path, even though there is no bound on the times of the requested updates. Denote its limit by L_∞. The common deterministic bound permits passing expectations through the limit. Indeed, for every $\varepsilon > 0$,

$$\mathbb{E}|L_k - L_\infty| \le \varepsilon + 2p_0^{-|\mathcal{S}\times\mathcal{A}|N} \mathbb{P}(|L_k - L_\infty| > \varepsilon),$$

and the probability tends to zero. Letting ε decrease to zero gives convergence in mean. For every event A known from $\mathcal{F}'_t$, the martingale identity consequently gives

$$\mathbb{E}[\mathbf{1}_A L_\infty] = \mathbb{P}(A).$$

These identities mean that $\mathbb{E}[L_\infty \mid \mathcal{F}'_t] = 1$: a conditional expectation is the quantity determined by the conditioning history that reproduces the original average on each event known from that history. The same argument at later histories gives $\mathbb{E}[L_\infty \mid \mathcal{F}'_k] = L_k$.

Define the auxiliary law by

$$\widehat{\mathbb{P}}(E) = \mathbb{E}[L_\infty \mathbf{1}_E].$$

It assigns more weight to runs with successful seed updates. A failed request makes $L_\infty = 0$, so all requests succeed under this law. An event of original probability zero still has probability zero after reweighting. For a random start t, use weight one before t, on $\{t = \infty\}$ and when $t \geq \zeta_r$, and combine the fixed-start constructions on $\{t = j\}$ over integers j. Each of these events is known at time j, so the same conditional argument applies.

It remains to check that future requests do not bias updates at completed states. If a bounded random variable Y is determined by the next history $\mathcal{F}'_{k+1}$, then on $\{L_k > 0\}$,

$$\widehat{\mathbb{E}}[Y \mid \mathcal{F}'_k] = \frac{\mathbb{E}[L_\infty Y \mid \mathcal{F}'_k]}{L_k} = \frac{\mathbb{E}[L_{k+1} Y \mid \mathcal{F}'_k]}{L_k}. \tag{55}$$

For the first equality, test the proposed conditional expectation against any event A known at k. The definition of the weighted law and $\mathbb{E}[L_\infty \mid \mathcal{F}'_k] = L_k$ give

$$\widehat{\mathbb{E}}\left[\mathbf{1}_A \frac{\mathbb{E}[L_\infty Y \mid \mathcal{F}'_k]}{L_k}\right] = \mathbb{E}[\mathbf{1}_A L_\infty Y] = \widehat{\mathbb{E}}[\mathbf{1}_A Y].$$

The fraction is defined as zero where $L_k = 0$; nonnegativity and the conditional-mean identity imply that those histories have auxiliary probability zero. The second equality in (55) uses $\mathbb{E}[L_\infty \mid \mathcal{F}'_{k+1}] = L_{k+1}$, so all later request factors average out, even when the later requests depend on the present return.

The current factor L_{k+1}/L_k has conditional mean one given the history and selected state. Applying (55) to the indicator that a particular state is selected therefore preserves state-selection probabilities. If that state is complete, the current factor is exactly one. Conditioning further on its selection consequently leaves the action and return law unchanged. The same holds after the construction stops. The algorithmic identities and the inertia rule also remain valid, since originally null events remain null. Finally,

$$\widehat{\mathbb{P}}(E \mid \mathcal{F}'_t) = \mathbb{E}[L_\infty \mathbf{1}_E \mid \mathcal{F}'_t] \leq p_0^{-|\mathcal{S}\times\mathcal{A}|N} \mathbb{P}(E \mid \mathcal{F}'_t),$$

which is (50). This reweighting is only a proof device; the algorithm continues to sample its original returns. ■

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